Product Specification

Table # Product Specification

Product Specification	
Category	**Specification**
Product Name	Sentrion SDS200
Product Type	CCTV-Based Smoke Detection and Automated Alert System
Description	Sentrion SDS200 is a CCTV-based smoke detection and automated alert system designed to enhance early fire hazard detection in indoor environments. It utilizes real-time video feeds and image processing techniques to identify visible smoke and trigger alerts. The system improves response time and provides a practical safety solution using existing CCTV infrastructure.
Detection Method	Vision-based smoke detection using image processing (OpenCV)
Input Source	IP Camera / USB Camera (CCTV integration)
Processing Unit	Raspberry Pi 4 Model B
Software Platform	Python with OpenCV library
Detection Features	Motion, color, and smoke pattern analysis with 6multi-frame verification

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Alert System	Buzzer (audible alarm) and LED indicator
Notification Method	Local network alert (Wi-Fi); optional remote notification (mobile/email/web)
Connectivity	Wi-Fi / Local Network
Power Supply	5V Power Adapter
Backup Power	Optional battery backup
Data Storage	MicroSD Card
Recorded Data	Event logs, timestamps, and snapshot images
Processing Type	Real-time video frame processing
Detection Response Time	Within a few seconds after confirmed detection
Monitoring Capability	Continuous (24/7 operation)
Enclosure	A6666666BS casing
System Design	Modular configuration for easy maintenance and upgrade
Application	Indoor environments (offices, retail stores, commercial spaces)